HEALTH-BAND Neuro: Product Roadmap

This document outlines the development trajectory for the HEALTH-BAND Neuro, detailing the progression from the V1 hardware to future iterations incorporating advanced display technologies and expanded sensor capabilities.

Phase 1: V1 Hardware (Current)

Status: Design Complete, Provisional Patent Filed

The V1 hardware establishes the foundational Zero-Hole Architecture and the core sensor suite.

- **Architecture:** Dual-purpose USB-C clasp mechanism (Zero-Hole design).
- **Display:** 0.49-inch Micro OLED (flush-mounted).
- **Storage:** 64GB NAND Flash (USB MSC capability).
- **Power:** 200mAh Li-Po battery with pass-through charging.
- **Sensors:** HR/SpO2, ECG, BAC (Fuel Cell), VOC, Skin Temp, IMU, UV.
- **Neuromodulation:** sEMG gesture input and TENS therapeutic output via 6x platinum electrodes.
- **Connectivity:** BLE 5.3 and USB 2.0 FS.

Phase 2: Firmware and Algorithm Refinement (Next 6-12 Months)

Status: In Progress

Focus shifts to optimizing the software stack running on the nRF52840 to fully utilize the V1 hardware capabilities.

- **sEMG Machine Learning:** Train on-device TinyML models to accurately classify complex hand gestures (e.g., pinch, swipe, fist clench) for UI navigation without touching the display.
- **TENS Protocol Development:** Develop specific electrical stimulation profiles for pain management, muscle recovery, and potentially tactile feedback (haptics via electrical stimulation).
- **BAC Calibration:** Refine the Venturi channel airflow dynamics and sensor calibration algorithms for the Dart fuel cell to ensure law-enforcement-grade accuracy.

Phase 3: V2 Hardware - The Holographic CIP (18-24 Months)

Status: Conceptual, CIP Patent Strategy Defined

The V2 iteration will focus on replacing the conventional OLED display with the technology outlined in the Continuation-In-Part (CIP) patent claim.

- **Display Evolution:** Transition from the rigid Micro OLED panel to a flexible micro-LED array positioned behind a holographic diffuser film.
- **Aesthetic Impact:** Achieve a truly bezel-less design where the display metrics appear to float on the surface of the carbon fiber band.
- **Structural Impact:** Improved durability and flexibility by eliminating the glass/rigid substrate of the OLED panel.
- **Sensor Expansion:** Investigate the integration of non-invasive continuous glucose monitoring (CBM) via advanced optical or transdermal techniques.

Phase 4: V3 Hardware - Clinical Integration (36+ Months)

Status: Future Planning

The V3 iteration aims to position the HEALTH-BAND Neuro as a certified medical device rather than a consumer health tracker.

- **FDA/CE Certification:** Pursue formal regulatory approval for the ECG, BAC, and TENS functionalities.
- **Closed-Loop Therapy:** Implement closed-loop systems where biometric data (e.g., tremor detection via IMU or stress markers via HRV/GSR) automatically triggers specific TENS/neuromodulation protocols.
- **Material Science:** Explore advanced biocompatible materials for the band and electrodes to support permanent, multi-week continuous wear.